Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/03/2023
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	02/07/2019	**End Date:**	02/07/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/7/2033 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/18/21) NOTE: End date changed to 6/6/2021 per NSSC information (Ed., 11/9/20)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI): In the fall of 2023, Co-Investigator Amir Kheradmand, M.D. (a neurologist colleague at Johns Hopkins University/JHU) was added to the project at the behest of the NASA Johnson Space Center (JSC) Neurosciences Lab. He will assist JSC in the analysis of video data on ocular torsion. [Ed., 1/18/24]		
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University) Kheradmand, Amir M.D. (Johns Hopkins University School of Medicine)		
Grant/Contract No.:	80NSSC19K0487		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose a simple and rapid measure of otolith asymmetry -- a basic low-level neurovestibular function that is altered in spaceflight. Our assessment apparatus and procedure was developed with previous Human Research Program (HRP) support, and has undergone validation testing and is currently in use in an unrelated Department of Defense (DoD) study. It provides a measure of binocular alignment (vertical and torsional), which is a reflection of underlying otolith asymmetry. While this measure might be important in itself (since it reflects a form of central neural compensation that might be challenged in spaceflight), it could also provide a simple alternative to locomotion and posture testing if it is found to correlate with those measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A simple and rapid assessment of neural/neurovestibular function, as described here, will be a useful in-flight procedure to track changes over the course of a mission. Given the widespread connection of neural circuits and functions with other body functions, it is not inconceivable that neurovestibular effects as assessed here will likewise have implications for a wide variety of neural, cognitive, and performance aspects of crews in flight. Our testing provides an opportunity to assess a low-level neural system that is not under volitional control, is responsive to g-level alterations, and might be an indicator of broader and more functionally relevant effects. The technology and procedures for this project are being used in DoD projects for the assessment of vestibular function in military personnel.
Task Progress:	Progress since November 2022 This project was absorbed into the CIPHER (Complement of Integrated Protocols for Human Research) Vestibular Health IRB (Institutional Review Board) project as part of a broader re-scoping of CIPHER early in 2022. Online integrated consent briefings were supported, and a presentation was made at the online HRP (Human Research Program) Investigators Workshop in February 2023. In the fall of 2023, Co-Investigator Amir Kheradmand, M.D. (a neurologist colleague at Johns Hopkins University/JHU) was added to the project at the behest of the NASA Johnson Space Center (JSC) Neurosciences Lab. He will assist JSC in the analysis of video data on ocular torsion. In the fall of 2021, we started a set of experiments in the lab to provide further baseline data by which to evaluate the upcoming flight data from this project. In these trials, subjects are evaluated for ocular alignment (OA) while standing on a hard floor, standing on foam, while seated, and while making head movements. This will help us evaluate the impact of postural demands on ocular alignment; we conjecture that the heightened demand while standing will call for a more precise compensation for any latent otolith asymmetry, causing a reduction in misalignment in that condition compared to seated. Six subjects were tested on an early version of the protocol (30-40 planned in total), which was later modified to include additional conditions relevant to otolith effects on posture and locomotion. A new student team is now working on this project, IRB approvals and training have been completed, and experiments will resume in 2024. A portion of the protocol for this project was modified and extended for use as an educational demonstration at Space Center Houston. This enables collection of a large set of normative data (N=167 currently) with which to refine our analyses, and which can provide context for interpretation of results from astronauts. Initial analysis of these data was performed by medical student Nabila Ali for an epidemiology degree project. This analysis so far shows a few things. Compliance was high in this volunteer population, with a 92% completion rate. Increasing age was significantly associated with increasing absolute median vertical ($b = 0.002$, $p = 0.001$) and torsional ($b = 0.006$, $p = 0.04$) ocular misalignment. Concussion history was associated with increased absolute median vertical (0.088, $p = 0.09$) and torsional ($b = 0.280$, $p = 0.16$) misalignment. Diabetes was associated with increased absolute torsional ($b = 0.409$, $p = 0.06$) misalignment. Smoking was significantly associated with decreased absolute torsional ($b = -0.369$, $p = 0.03$) misalignment. Final multivariate models for vertical misalignment included age and concussion history and torsional misalignment included age, concussion history, diabetes, and smoking. The connection to smoking is particularly interesting and suggests a connection to nicotine as a relevant parameter. This project also includes a component dedicated to data-integration of the other CIPHER projects. While awaiting CIPHER flight data, we have been performing several preparatory activities so that we can be better prepared to organize and analyze that data: 1. An updated set of hypotheses (e.g., expected cross-study associations or correlations) and methods is being prepared, per the Statement of Work for this project. This will be provided to HRP in early 2024. This update will be based on the revised (final) schedule of data acquisition, and the revised (final) set of studies. It may require a reexamination of the text analytics previously performed (see below) and another set of such analyses based on the revised plan. 2. A manuscript describing the scientific background and justification for CIPHER has been prepared by the Principal Investigator (PI). Assistance in outlining the content was provided by Kristin Fabre of HRP, and further discussions on content were held with Cherie Oubre. This substantial report will require coordination with the journal where it is submitted, since its length (currently 6,666 words) pushes the limits for such a manuscript, and descriptions of each individual CIPHER study have not yet been incorporated. Discussion of this manuscript will take place at Investigators' Workshop (IWS) 2024. 3. Discussions have begun with several groups on how to acquire, archive, and curate the data that are expected to be provided to the PI by the other CIPHER investigators. This includes discussion with personnel from the ALSDA (Ames Life Sciences Data Archive) and their Open Science Data Repository (OSDR) project, which has created a set of templates for investigators to provide critical background data on each experiment as they upload data (nominally from the Space Biology program). Further conversations with LSDA identified a similar set of documents and guidelines that are in place for HRP data and LSDA. Work continues with LSDA to determine similarities and differences between these approaches, if ideas from the two approaches might be combined for the needs of CIPHER, and how examination of the HRP DAGs (Directed Acyclic Graphs) (see below) might inform updates to the data-ingestion templates (for example by identifying important but overlooked metadata). 4. Other discussions with LSDA continue in order to identify datasets that are already available, that could help address some of the same questions as CIPHER (i.e., are from the same or similar body systems), and would permit longitudinal (personalized) investigation. Potential datasets include published data from the early space program up through Skylab (when data were published in identifiable/attributable form), and LSDA data from STS (Space Transportation System) missions. While longitudinal (personalized) data are highly preferred (since that is what will be provided by CIPHER),

	pooled/population data might be useful in establishing bounds and broad cross-disciplinary correlations. (Graduate student Alyssa Columbus will work on this task with her academic advisors in biostatistics.) 5. Through discussions with Anand Narayanan, we have identified a Space Biology project (Effects of Simulated Microgravity and Partial Unloading on Organ Systems of the Body, PI: Michael Delp) that will acquire large amounts of multi-system data from animal models. Since this project will acquire and analyze significant data before CIPHER, it can help provide guidance as to some relevant cross-discipline analysis methods. The HRP DAGs (see below) can provide direction in the cross-fertilization of these two projects (Reynolds RJ, Scott RT, Turner RT, Iwaniec UT, Bouxsein ML, Sanders LM, Antonsen EL. Validating causal diagrams of human health risks for spaceflight: an example using bone data from rodents. Biomedicines. 2022 Sep 5;10(9):2187). 6. Plans for the next grant period also include: a. Repetition of text analytics on the final configuration of CIPHER proposals, to identify common themes, methods, or approaches across studies. As before, this will involve examination of reference lists in each CIPHER proposal for citations to common authors, and Latent Dirichlet Allocation (LDA) to identify underlying concepts that might connect different proposals. This will inform the generation of new hypotheses (above). b. Mapping of DAGs to the CIPHER studies, as another way to examine common areas across studies. DAGs have been created for all of the HRP Risks (Antonsen EL, et al. Directed Acyclic Graphs: A Tool for Understanding the NASA Human Spaceflight System Risks - NASA/TP-20220015708) and provide a common framework for discussing cross-disciplinary interactions and relations. c. Development of a conceptual model that might identify astronaut subjects as “good” or “bad” adapters through broad general analyses of datasets, looking for groupings of parameters that can be ranked in terms of highly adaptive or less adaptive (ordinal rather than numerical analysis). (Proposed by undergraduate student Iuliia Dmitrieva.)
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Macaulay TR, Wood SJ, Bollinger A, Schubert MC, Shelhamer M, Bishop MO, Reschke MF, Clément G. "Comparison of asymmetry between perceptual, ocular, and postural vestibular screening tests." Brain Sciences. 2023 Jan 23;13(2):189. https://doi.org/10.3390/brainsci13020189 ; PubMed PMID: 36831732; PubMed Central PMCID: PMC9954277 , Jan-2023
Articles in Peer-reviewed Journals	Shelhamer M, Antonsen E, Reynolds R. "The need for a systems approach in spaceflight health and medicine." IEEE Xplore. 2023 May 15. https://doi.org/10.1109/aero55745.2023.10115773 , May-2023
Articles in Peer-reviewed Journals	Shelhamer M, Gersh JR. "A mission architecture to integrate human and spacecraft functional performance." IEEE Xplore. 2023 May 15:1-8. https://doi.org/10.1109/aero55745.2023.10115715 , May-2023